INCH-POUND

MIL-S-40626D(GL)
31 March 1994
SUPERSEDING
MIL-S-40626C(GL)
7 March 1978

MILITARY SPECIFICATION

SIGN KIT, VEHICLE CLASS

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of vehicle class sign kit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898 - Cushioning Material, Cellulosic, Packaging

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5017, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9905

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QQ-A-225/3 - Aluminum Alloy Bar, and Wire; Rolled, Drawn, or Cold Finished, 2011
QQ-A-250/8 - Aluminum Alloy 5052, Plate and Sheet
QQ-S-698 - Steel Sheet and Strip, Low Carbon
TT-C-490 - Cleaning Methods for Ferrous Surfaces and Retreatments for Organic Coatings
TT-P-1757 - Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant

STANDARDS

FEDERAL

FED-STD-105 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

6-1-2248 - Sign Kit, Vehicle Class; Assembly and Detail

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-MII, Natick, MA 01760-5017.)

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2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Materials and components. The materials and components shall be as specified herein and as shown on the applicable drawing. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.2.1 Aluminum.

3.2.1.1 Plate and sheet. The aluminum alloy plate and sheet shall be as specified in QQ-A-250/8.

3.2.1.2 Bar, rod and wire. The aluminum alloy bar, rod and wire shall conform to temper T3 of QQ-A-225/3.

3.2.2 Steel, sheet and strip. The steel sheet shall conform to cold rolled or hot rolled, commercial quality, and the steel strip shall conform to any temper, condition and edge of QQ-S-698.

3.2.3 Cap screws, washers and nuts. Cap screws, washers and nuts shall conform to the requirements shown on Drawing 6-1-2248.

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3.2.4 Primer. The primer shall be as specified in TT-P-1757 with composition and color optional.

3.2.5 Coating. The coating shall be as specified in MIL-C-53039.

3.3 Construction. The sign kit shall conform in all respects to Drawing 6-1-2248 and as specified herein.

3.3.1 Disk. The disk shall be fabricated of aluminum alloy, temper O, as specified in 3.2.1.1.

3.3.2 Anchor pin and pin lock. The anchor pin and pin lock shall be fabricated of aluminum alloy specified in 3.2.1.2. The pins, after assembly, shall not be loose.

3.3.3 Plates, retaining and numeral. The retaining and numeral plates shall be fabricated of aluminum alloy, temper H38, as specified in 3.2.1.1. The numeral and retaining plates shall be inserted in position and shall not become loose or fall out due to vibration when tested as specified in 4.4.5. Prepainted aluminum sheet for numeral plates may be furnished and shall conform to the manufacturers standard commercial practice. The background color shall be Dark Green 34082, as specified in 3.2.5.

3.3.4 Brackets, universal, long and short. The brackets shall be fabricated of steel as specified in 3.2.2 and shall be of the design as shown on Drawing 6-1-2248.

3.4 Finish. Except when prepainted sheet is used, the sign kit shall be finished as specified herein.

3.4.1 Surface preparation. All surfaces and edges of the disk, retaining plate, and numeral plates shall be prepared for painting in accordance with class 1A of MIL-C-5541. The brackets shall be prepared for painting in accordance with any method of TT-C-490.

3.4.2 Priming. All prepared surfaces and edges of disks, retaining plate, and numeral plates shall be coated with the primer specified in 3.2.4.

3.4.3 Coating. After drying, the primed surfaces and edges of the disk, retaining plate, steel brackets, and numeral plates shall be coated as specified in 3.2.5.

3.4.4 Characters. The letter "C" and the numerals shall be gothic type and shall be applied to the retaining plate and numeral plates, respectively, after the background color specified in 3.3.3 has been applied. The letter "C" and the numerals shall be clear and sharply defined. Additional requirements for the characters shall be as specified in table I.

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TABLE I. Additional character requirements

Requirement	Letter	Numerals	
Height	2 inch minimum 2-5/8 inch maximum	3-1/2 inch minimum 3-5/8 inch maximum	
Width of stroke	7/16 inch minimum 9/16 inch maximum	9/16 inch minimum 11/16 inch maximum	
Color (see 3.2.5)	Aircraft Red 31136	Black 37030	
Data	Letter "C" centered on one side of the retaining plate only	One side	Opposite side
		0	1
		2	3
		4	5
		6	7
		8	9

3.5 Marking for identification. Each sign kit disk shall be permanently and legibly marked on the back side with the manufacturer's name or readily identifiable trademark.

3.6 Workmanship. The end item shall conform to the quality of product established by this specification. All edges shall be rounded or beveled and all surfaces and edges shall be free from burrs, slivers and sharp edges. The finish of the disks, retaining plate, numeral plates, brackets, lettering and numerals shall be free from imbedded foreign matter, uncoated areas, cracks, blisters, flakes, scratches, mars, sags and runs. The lettering shall be smooth, even, straight and perpendicular to the base.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

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ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.5.1. Any nonconformance shall be cause for the rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. When applicable, inspection shall be made to determine that surface preparation and application of primer to the components are in accordance with specified requirements of 3.4.1 and 3.4.2. Whenever nonconformance is noted, correction shall be made to the affected items and process. Items that cannot be corrected shall be removed from the process.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of sign kits. The sample unit shall be one sign kit. The inspection level shall be I. The finding of any defect shall be cause for the rejection of the lot.

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TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish (all components)	Component not colors specified	101	
	Cap screws, nuts and washers not zinc plated		201
	Coated surfaces not smooth, uniform or continuous		202
	Uncoated areas	102	
	Not free from cracks, blisters, orange peel, flakes, sags, runs, scratches or mars		203
	Foreign matter imbedded		204
	Touch up, when applicable, not uniformly accomplished		205
Design	Component parts do not assemble together or fit snugly	103	
Construction and workmanship	Any hole missing	104	
	Embossments missing or not located as shown on drawing	105	
	Bracket and disk holes not in alignment	106	
	Pins not securely riveted	107	
	Rivet heads not countersunk as shown		206
	Characters not gothic type	108	
	Letter "C" not applied as specified		207
	Numerals not applied on both surfaces of numeral plates	109	
	Numeral plates not numbered as specified	110	
	Not required number of numeral plates	111	
	Characters not centered		208
	Sharp edges, burrs or slivers not removed	112	
	Edges and corners not rounded		209
	Edge of letter or numerals not clearly defined		210
Identification marking	Missing, incomplete, not legible		211

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on Drawing 6-1-2248 and as specified herein. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not

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within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of sign kits. The sample unit shall be one sign kit. The inspection level shall be S-4. The finding of any defect shall be cause for rejection of the lot.

4.4.5 End item testing. The end items shall be tested for the vibration characteristic listed in 4.5.1. The lot size shall be expressed in units of sign kits. The sample unit shall be one sign kit. The inspection level shall be S-3. Any test failure shall be cause for rejection of the lot.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2. The finding of any defect shall be cause for rejection of the lot.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1. The finding of any defect shall be cause for rejection of the lot.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Vibration test. The sign kit assembly, with brackets attached to simulate mounting on a vehicle in an upright position, shall be affixed to any commercial vibration testing machine, and tested for compliance with 3.3.3. The vibration test shall be conducted for one hour at a frequency rate of 1000 cycles per minute at an amplitude of 1/4 inch. Any numeral, letter, or retaining plate falling out due to vibration or found to be loose after the vibration test shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. The numeral plates, retaining plate and long bracket shall be assembled onto the disk with the bracket secured in place with cap screws, lockwashers and nuts as shown on Drawing 6-1-2248. The four cap screws, four lockwashers and four nuts for the short bracket, and the two 1-3/4 inch long cap screws (items 11, 12, 13, and 14) on Drawing 6-1-2248 shall be placed in a plastic bag and the bag closed by heat sealing, tape or similar means. The assembled sign kit shall be wrapped in 3/4 inch nominal thickness cushioning material conforming to grade II, class B of A-A-1898. The wrapped sign kit shall then be packaged in a paperboard box. The cushioned disk shall be placed on the bottom of the box with the numeral plates face down. The short bracket and the plastic bag containing cap screws, lockwashers and nuts shall then be placed in the box and all voids filled with cushioning material specified herein. Closure of the box shall be in accordance with the appendix of the applicable box specification.

5.1.2 Commercial preservation. Sign kits shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twenty-four sign kits preserved as specified in 5.1, shall be packed in a snug-fitting, fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. The sign kits shall be packed flat, two in length, two in width, and six in depth. Each shipping container shall be closed in accordance with method III, water-proofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing. Twenty-four sign kits, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. The sign kits shall be packed flat, two in length, two in width, and six in depth. Closure shall be in accordance with method II of the appendix of PPP-B-636.

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5.2.3 Commercial packing. Sign kits, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), sign kits, packed as specified in 5.2.2 and 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The vehicle class sign kit is used to afford immediate recognition of the weight classification of each vehicle and as an aid in traffic control at bridges and crossings.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).
e. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement as cited in NATO STANAG No. 2010. When amendment, revision, or cancellation of this specification is proposed that will affect or violate the international

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agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.5 Subject term (key word) listing.

Identification marking
Traffic control
Vehicle weight
Weight classification

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - ME, MD
DLA - GS

User activities:

Army - AT
Navy - MC, YD

Civil agency coordinating activity:

GSA - FSS

Preparing activity:

Army - GL

(Project 9905-A320)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:

1. DOCUMENT NUMBER
MIL-S-40626D(GL)

2. DOCUMENT DATE *(YYMMDD)*
940331

3. DOCUMENT TITLE
Sign Kit, Vehicle Class

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*

b. ORGANIZATION

c. ADDRESS *(Include Zip Code)*

d TELEPHONE *(Include Area Code)*
(1) Commercial
(2) AUTOVON *(If applicable)*

7. DATE SUBMITTED *(YYMMDD)*

8 PREPARING ACTIVITY

a NAME
U.S. Army Natick RD&E Center

b TELEPHONE *(Include Area Code)*
(1) Commercial: 508-651-5235
(2) AUTOVON/DSN: 256-5235

c ADDRESS *(Include Zip Code)*
Commander, U.S. Army Natick RD&E Center
ATTN: SATNC-UC
Natick, MA 01760-5017

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DD Form 1426, OCT 89 NATICK OP3, JUL 91 *Previous editions are obsolete*

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